

Report
Of
Data Analytics with Python & Power BI

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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Batch: 2026-27

2026-27

CANDIDATE'S DECLARATION

I, Piyush Kumar, do hereby declare that the internship report titled "Data Analytics with Python & Power BI" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Piyush Kumar

Roll No.: 2410030021

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Piyush Kumar

Roll No.: 2410030021

INTERNSHIP COMPLETION CERTIFICATE

Internship Certificate

Certificate issued by EduSkills Academy

CERTIFICATE
OF INTERNSHIP

PRESENTED TO

Piyush Kumar

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

Data Analytics with Python & Power BI

During the internship, the learner demonstrated proficiency in:

- Welcome to Data Analytics & Environment Setup & Core Python for Data Analysis
- Data Cleaning & Preprocessing with Pandas & SQL Fundamentals for Data Analysis
- Advanced SQL Techniques & Data Extraction & Data Visualization with Power BI - Basics
- Python & SQL Integration for Data Extraction & Power BI Data Modeling & DAX Basics
- Advanced Power BI Visualizations & Interactivity & Python for Data Visualization
- Python & Power BI Integration & Power BI Service & Collaboration
- Real-World Project: Sales Performance Analysis & Real-World Project: Customer Behavior Analysis & Best Practices & Project Workflow
- Capstone Project: E-commerce Sales & Customer Insights Dashboard

This certification recognizes the successful completion of the internship requirements.

Issued on: Jul 27, 2020
Certificate ID: 2020-2PC28F17AA

Mihir Swain
General Manager
(Learning & Development)
EduSkills

TABLE OF CONTENTS

Chapter No.	Chapter Name	Page Nos.
1.	Candidate's Declaration	i
2.	Acknowledgment	ii
3.	Internship Completion Certificate	—
4.	Project Description	1
4.1	Introduction	1
4.2	Organization Profile	2
4.3	Problem Statement	2
4.4	Project Objectives	3
4.5	Scope of the Project	3
4.6	Technologies and Tools Used	4
4.7	System Architecture	5
4.8	Methodology	5
4.9	Expected Outcomes	6
4.10	Certificates of completion and communication proof	6
5.	Bibliography / References	7

LIST OF TABLES

Table No.	Title	Page
4.1	Technologies and Tools Used	4
7.1	Capstone Dashboard Metrics	7
8.1	Key Skills and Outcomes	8

LIST OF FIGURES

Figure No.	Title	Page
4.1	Integrated Analytics Workflow	5

A.1	Internship Completion Certificate	9
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4. PROJECT DESCRIPTION

4.1 Introduction

The internship titled “Data Analytics with Python & Power BI” was an 8-week structured learning journey in data analytics. The program focused on building practical capabilities in Python, Pandas, SQL, Power BI and data visualization. The learning path moved from environment setup and core programming concepts to data preparation, analytical querying, business intelligence, visualization and integrated project workflows.

The internship also provided practical exposure through sales performance analysis, customer behavior analysis and an end-to-end capstone project. The capstone, titled “E-commerce Sales & Customer Insights Dashboard,” brought together analytics, visualization and dashboarding skills and focused on sales and customer insights in an e-commerce context.

4.2 Organization Profile

EduSkills Academy is identified in the supplied internship presentation as the organization for the 8-week internship and in the certificate as the certificate issuer. The documented internship domain was Data Analytics with Python & Power BI. The available materials describe the program through its learning modules, practical projects, integrated workflow and capstone dashboard.

The supplied materials do not provide additional organizational history or corporate details. Accordingly, this report keeps the organization profile limited to the information documented in the internship presentation and certificate.

4.3 Problem Statement

The practical problem addressed during the internship was the transformation of raw and structured data into analysis-ready information and meaningful business insights. Data must be prepared, queried, analyzed and communicated effectively before it can support interpretation and reporting.

The internship addressed this workflow through Python-based analytical programming, Pandas-based cleaning and preprocessing, SQL-based extraction and analysis, and Power BI-based modeling, DAX and interactive visualization. The

project work applied these capabilities to sales, customer behavior and e-commerce analytics.

4.4 Project Objectives

- To develop foundations in Python for data analysis and visualization.
- To learn data cleaning and preprocessing using Pandas.
- To develop SQL fundamentals, advanced SQL techniques and data extraction skills.
- To understand Power BI basics, data modeling, DAX and advanced visualizations.
- To integrate Python, SQL and Power BI into an end-to-end analytics workflow.
- To apply analytics concepts through sales performance and customer behavior projects.
- To complete an e-commerce sales and customer insights dashboard as the capstone project.
- To improve the ability to communicate analytical insights through visual reports and dashboards.

4.5 Scope of the Project

The scope of the internship covered the complete learning workflow from analytics environment setup to analysis and reporting. It included Python for data extraction, analytical programming and visualization; Pandas for data cleaning and preprocessing; SQL for structured querying and data extraction; and Power BI for data modeling, DAX, interactive reporting, dashboard design and collaboration.

The practical scope included sales performance analysis, customer behavior analysis, structured project workflow practice and the capstone E-commerce Sales & Customer Insights Dashboard. Python–SQL and Python–Power BI integration were also part of the documented learning experience.

4.6 Technologies and Tools Used

Technology / Tool	Area of Use	Skills Developed
Python	Data analysis and visualization	Environment setup, core Python, analytical programming and visualization
Pandas	Data cleaning and	Cleaning and

	preprocessing	preprocessing workflows
SQL	Structured data analysis	SQL fundamentals, advanced techniques and data extraction
Power BI	Business intelligence	Visualization, data modeling, DAX, dashboards, Service and collaboration
Python + SQL	Integrated workflow	Practical data extraction and analytical workflows
Python + Power BI	Integrated visualization	Integration of Python analysis with Power BI reporting

4.7 System Architecture

The documented integrated analytics workflow connects the major tools according to their roles in the analysis process.

Stage	Tool	Primary Activity
1	Python	Prepare, explore and analyze data
2	Pandas	Clean and preprocess data
3	SQL	Extract and query structured data
4	Power BI	Model data and create interactive visuals
5	Analytics Output	Communicate insights through reports and dashboards

Figure 4.1: Data → Prepare & Explore → Extract & Query → Model & Visualize → Communicate Insights

4.8 Methodology

- **Environment Setup:** Set up the working environment and developed core Python concepts required for analytics.

- Data Preparation: Used Pandas-based cleaning and preprocessing workflows to prepare reliable, analysis-ready information.
- SQL Analytics: Practiced SQL fundamentals, advanced SQL techniques and data extraction for structured data.
- Exploration and Visualization: Used Python for analytical exploration and data visualization.
- Business Intelligence: Applied Power BI for modeling, DAX, interactive visualization, reporting and dashboard design.
- Integrated Workflow: Connected Python, SQL and Power BI into a practical analytics workflow.
- Project Practice: Applied the learning through sales performance analysis, customer behavior analysis and structured project workflow.
- Capstone: Combined the learned capabilities in the E-commerce Sales & Customer Insights Dashboard.

4.9 Expected Outcomes

The internship materials document the following learning outcomes and demonstrated capabilities:

- Core Python for data analysis and visualization.
- Data cleaning and preprocessing with Pandas.
- SQL fundamentals, advanced techniques and data extraction.
- Power BI visualization, modeling, DAX and Power BI Service.
- Integrated Python + SQL + Power BI workflows.
- Practical analytics projects covering sales, customer behavior and e-commerce dashboarding.

4.10 Certificates of Completion and Other Communication Mail Proof

The internship completion certificate issued by EduSkills Academy is included in this report. The supplied presentation also documents the internship overview, learning modules, real-world projects and capstone project. No separate communication-mail proof was included in the supplied files; therefore, no unsupported email or communication record has been added.

5. BIBLIOGRAPHY / REFERENCES

The report has been prepared using the supplied internship report format, the supplied internship presentation and the internship completion certificate. The following references represent the source materials used for this report:

- Internship Presentation: Data Analytics with Python & Power BI, 8-Week Internship, Piyush Kumar, Enrollment No. 2410030021, EduSkills Academy / IILM University.
- Internship Completion Certificate issued by EduSkills Academy for the Data Analytics with Python & Power BI internship.
- Internship Report Format / Instructions supplied by IILM University, School of Computer Science and Engineering.

APPENDIX A – DETAILED INTERNSHIP LEARNING RECORD

A.1 Python for Data Analysis

The internship covered environment setup, core Python concepts for data analysis, Python for data extraction and analytical workflows, Python for data visualization, and integration of Python with Power BI.

A.2 Data Cleaning, Pandas & SQL

Pandas-based cleaning and preprocessing workflows were used for reliable analysis. SQL learning covered foundations, advanced techniques and data extraction. Python and SQL were integrated for practical data extraction workflows.

A.3 Power BI & Business Intelligence

Power BI learning covered visualization from basic to advanced levels, data modeling and DAX basics, interactive reporting and dashboard design, Power BI Service and collaboration, and Python & Power BI integration.

A.4 Visualization & Integrated Analytics

The integrated workflow documented in the presentation is: Python – Prepare & Explore; SQL – Extract & Query; Power BI – Model & Visualize; Insights – Communicate. The key practice was to use Python for analytical exploration and visualization, SQL for structured extraction and querying, and Power BI for modeling, interactive visuals and collaboration.

APPENDIX B – REAL-WORLD PROJECTS

B.1 Sales Performance Analysis

Applied data analytics concepts to study sales performance and generate business-oriented insights.

B.2 Customer Behavior Analysis

Worked with customer-oriented analytics to understand behavior patterns and support data-driven interpretation.

B.3 Project Workflow

Practiced best practices and a structured project workflow from data preparation through analysis and reporting.

B.4 Practical Integration

Combined Python, SQL and Power BI concepts into an end-to-end analytics learning experience.

APPENDIX C – CAPSTONE PROJECT

C.1 E-commerce Sales & Customer Insights Dashboard

The capstone project was completed as part of the internship and brings together analytics, visualization and dashboarding skills. It focuses on sales and customer insights in an e-commerce context.

Metric	Value shown in supplied presentation
Revenue	₹ 4.8M
Orders	18.2K
Customers	9.7K

Table 7.1: Capstone dashboard metrics shown in the internship presentation.

APPENDIX D – KEY SKILLS & OUTCOMES

Skill Area	Documented Capability
Python	Core Python for data analysis and visualization
Pandas	Data cleaning and preprocessing
SQL	Fundamentals, advanced techniques and

	extraction
Power BI	Visualization, modeling, DAX and Service
Integration	Python + SQL + Power BI workflows
Analytics Projects	Sales, customer behavior and e-commerce dashboard

APPENDIX E – INTERNSHIP CERTIFICATE

Internship Certificate

Certificate issued by EduSkills Academy

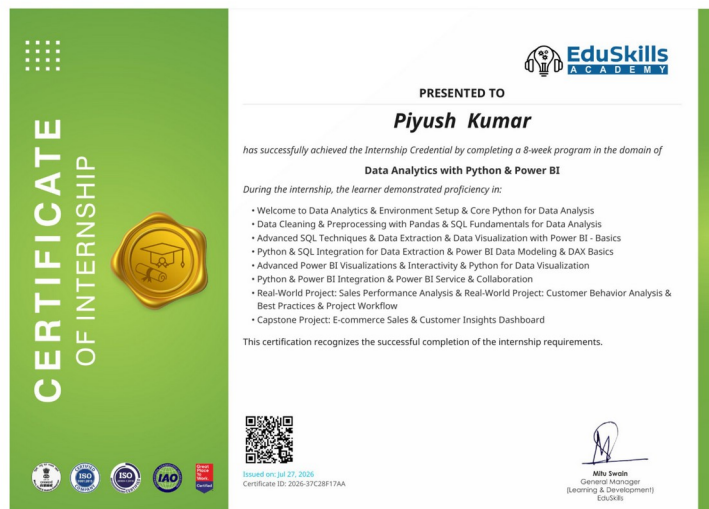


Figure A.1: Internship completion certificate issued by EduSkills Academy.